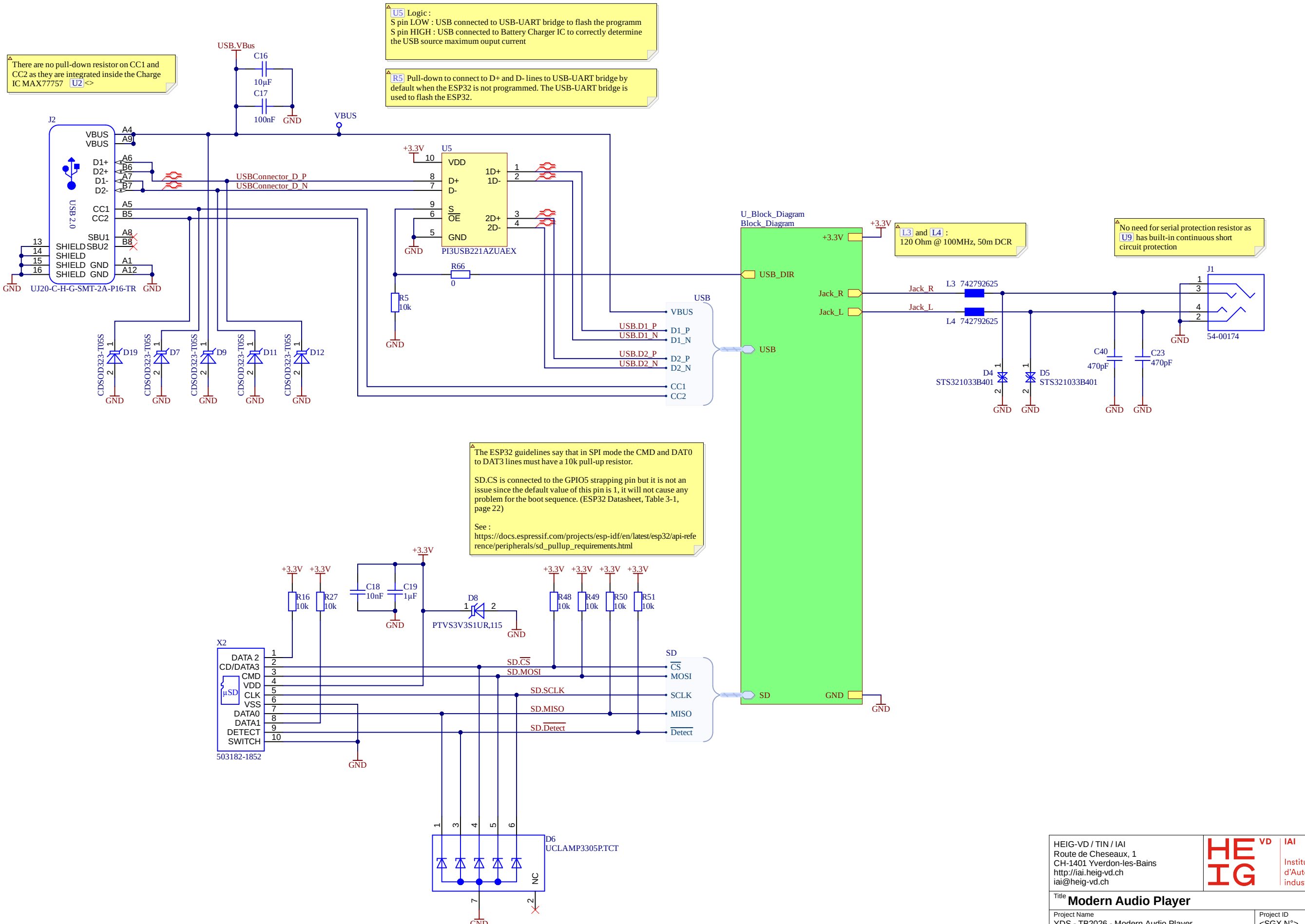
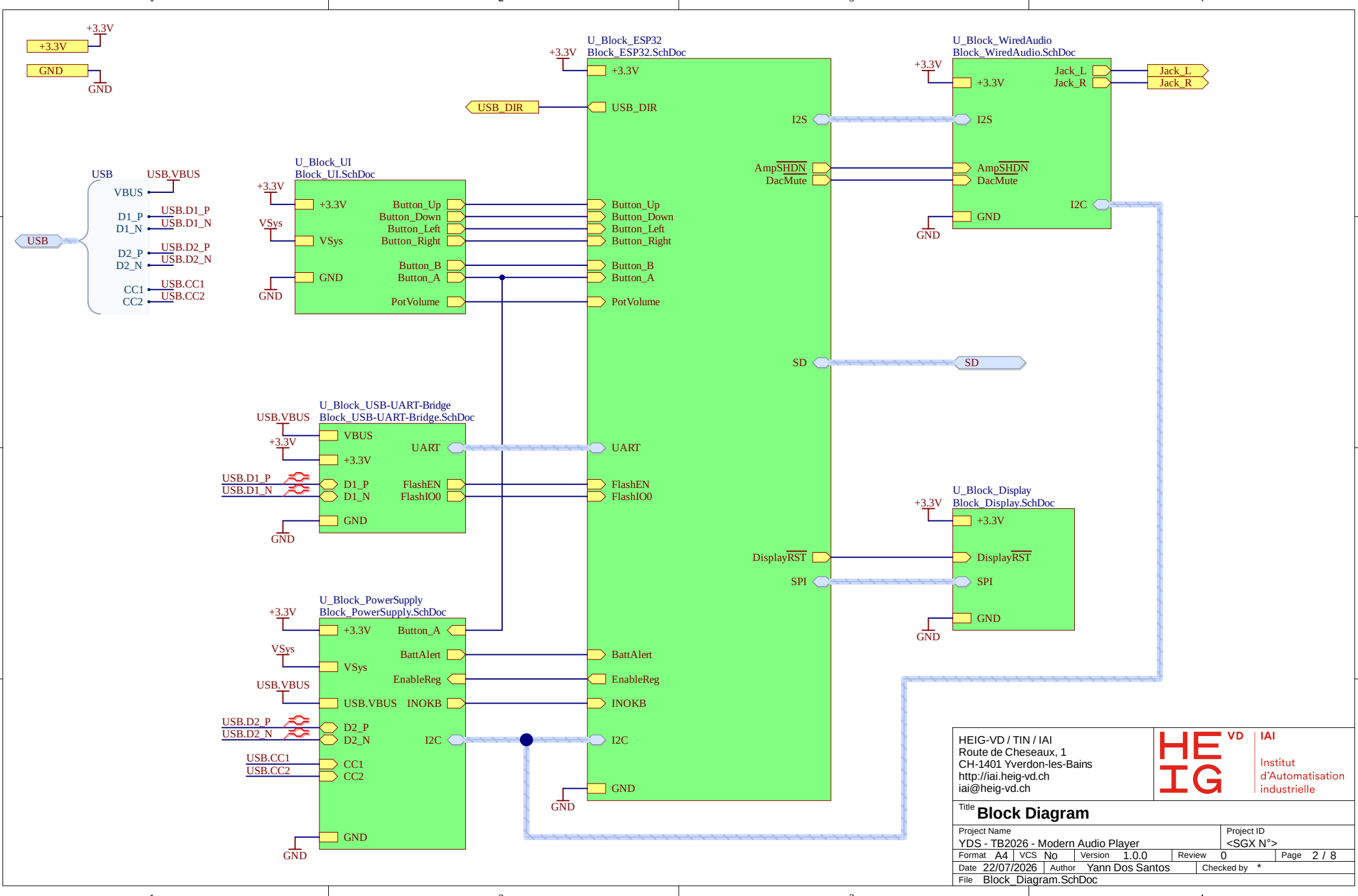


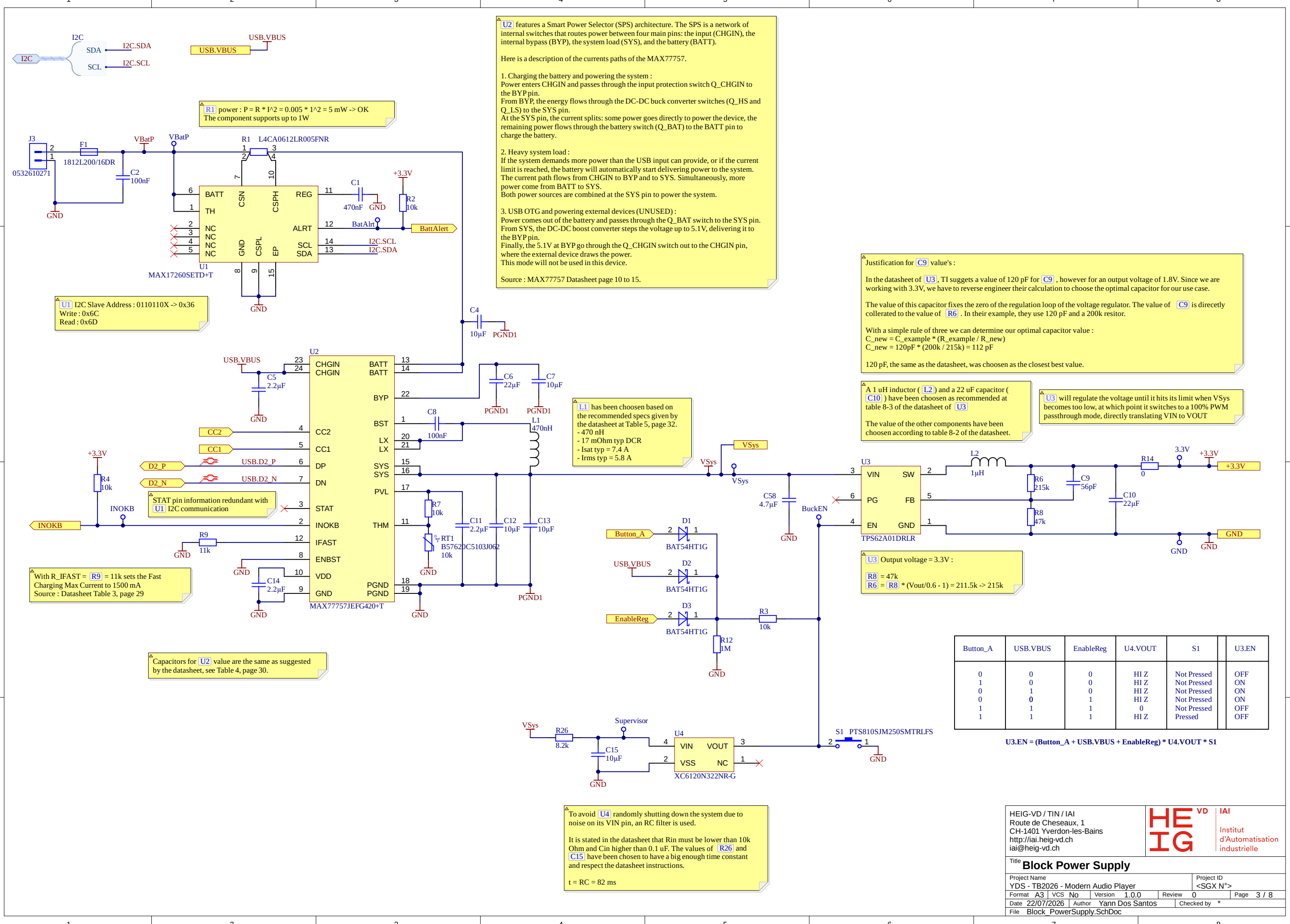




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Title Block Diagram						Project ID <SGX N°>	
Project Name YDS - TB2026 - Modern Audio Player							
Format	A4	VCS	No	Version	1.0.0	Review	0
Date	22/07/2026		Author	Yann Dos Santos		Checked by	*
File	Block_Diagram.SchDoc						



U2 features a Smart Power Selector (SPS) architecture. The SPS is a network of internal switches that routes power between four main pins: the input (CHGIN), the internal bypass (BYP), the system load (SYS), and the battery (BATT).

Here is a description of the currents paths of the MAX77757.

- Charging the battery and powering the system :**
Power enters CHGIN and passes through the input protection switch Q_CHGIN to the BYP pin.
From BYP, the energy flows through the DC-DC buck converter switches (Q_HS and Q_LS) to the SYS pin.
At the SYS pin, the current splits: some power goes directly to power the device, the remaining power flows through the battery switch (Q_BAT) to the BATT pin to charge the battery.
- Heavy system load :**
If the system demands more power than the USB input can provide, or if the current limit is reached, the battery will automatically start delivering power to the system.
The current path flows from CHGIN to BYP and to SYS. Simultaneously, more power come from BATT to SYS.
Both power sources are combined at the SYS pin to power the system.
- USB OTG and powering external devices (UNUSED) :**
Power comes out of the battery and passes through the Q_BAT switch to the SYS pin.
From SYS, the DC-DC boost converter steps the voltage up to 5.1V, delivering it to the BYP pin.
Finally, the 5.1V at BYP go through the Q_CHGIN switch out to the CHGIN pin, where the external device draws the power.
This mode will not be used in this device.

Source : MAX77757 Datasheet page 10 to 15.

Justification for **C9** value's :

In the datasheet of **U3**, TI suggests a value of 120 pF for **C9**, however for an output voltage of 1.8V. Since we are working with 3.3V, we have to reverse engineer their calculation to choose the optimal capacitor for our use case.

The value of this capacitor fixes the zero of the regulation loop of the voltage regulator. The value of **C9** is directly collatered to the value of **R6**. In their example, they use 120 pF and a 200k resistor.

With a simple rule of three we can determine our optimal capacitor value :
 $C_{new} = C_{example} \cdot (R_{example} / R_{new})$
 $C_{new} = 120pF \cdot (200k / 215k) = 112 \text{ pF}$

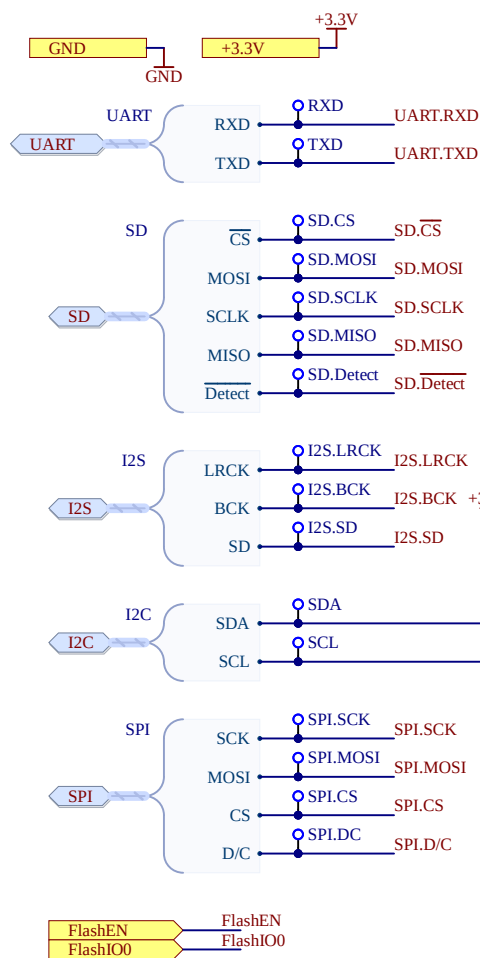
120 pF, the same as the datasheet, was chosen as the closest best value.

A 1 uH inductor (**L2**) and a 22 uF capacitor (**C10**) have been chosen as recommended at table 8-3 of the datasheet of **U3**

The value of the other components have been chosen according to table 8-2 of the datasheet.

U3 will regulate the voltage until it hits its limit when VSys becomes too low, at which point it switches to a 100% PWM passthrough mode, directly translating VIN to VOUT

Button_A	USB.VBUS	EnableReg	U4.VOUT	S1	U3.EN
0	0	0	HI Z	Not Pressed	OFF
1	0	0	HI Z	Not Pressed	ON
0	1	0	HI Z	Not Pressed	ON
0	0	1	HI Z	Not Pressed	ON
1	1	1	0	Not Pressed	OFF
1	1	1	HI Z	Pressed	OFF



△ I2C 400 kHz, pull-up resistor calculation.
Source : <https://www.ti.com/lit/an/slua689/slua689.pdf?ts=1778213985956>

$R_{p(\min)} = (V_{cc} - 0.4) / 0.003 = (3.3 - 0.4) / 0.003 = 967 \text{ Ohm}$
 $R_{p(\min)}$ is fixed no matter other parameters (since we only use +3.3V)

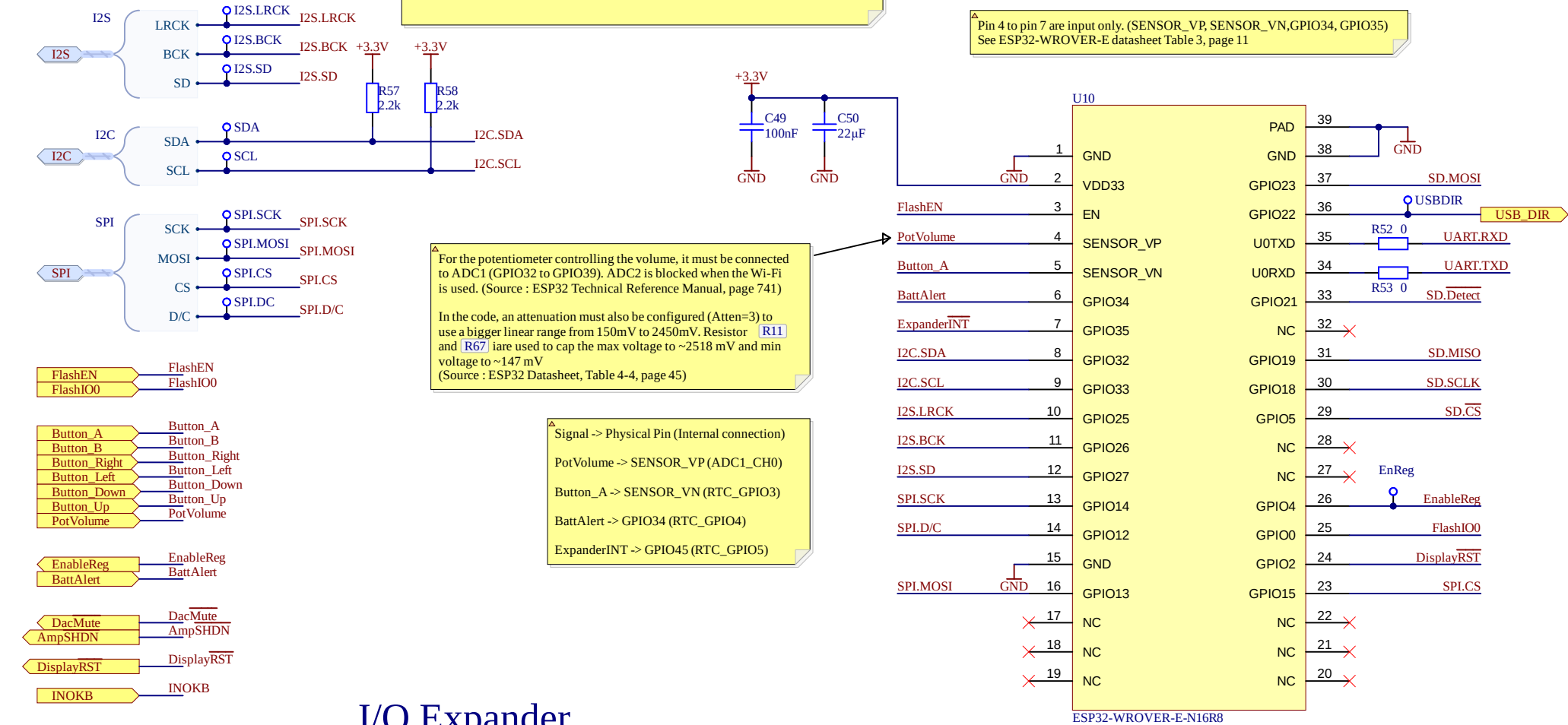
$R_p(\max) = t_r / (0.84 / 3 * C_b)$
 t_r is the maximum rising time depending of the frequency of the communication
 C_b is all the capacitance of the bus combined. To determine it, we add all the pin input capacitance of each device connected to the I2C bus, and then take a safety margin.

tr = 300 ns (see source Table 1)

To calculate C_b , we must add the capacitance of U10, U11, U8 and U1
 $C_b = 28 \text{ pF}$

Now we take a safety margin and calculate R (we will use C = 50 pF and C = 100 pF):
 $R_{p(max)1} = 300ns / (0.8473 * 50pF) = 7.08k \text{ Ohm}$
 $R_{p(max)2} = 300ns / (0.8473 * 100pF) = 3.54k \text{ Ohm}$

We these calculation, 2k2 is a good choice for the pull-ups resistors

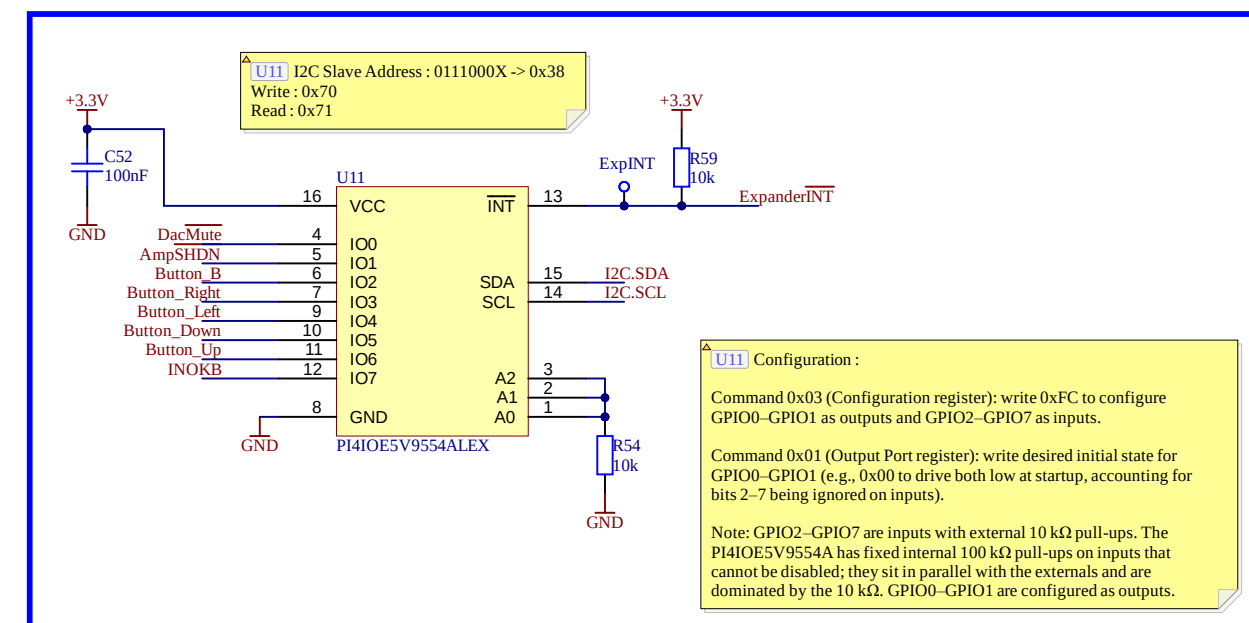


For the potentiometer controlling the volume, it must be connected to ADC1 (GPIO32 to GPIO39). ADC2 is blocked when the Wi-Fi is used. (Source : ESP32 Technical Reference Manual, page 741)

In the code, an attenuation must also be configured (Atten=3) to use a bigger linear range from 150mV to 2450mV. Resistor [R11](#) and [R67](#) are used to cap the max voltage to ~2518 mV and min voltage to ~147 mV (Source : ESP32 Datasheet, Table 4-4, page 45)

- Signal -> Physical Pin (Internal connection)
- PotVolume -> SENSOR_VP (ADC1_CH0)
- Button_A -> SENSOR_VN (RTC_GPIO3)
- BattAlert -> GPIO34 (RTC_GPIO4)
- ExpanderINT -> GPIO45 (RTC_GPIO5)

I/O Expander



U11 Configuration :

Command 0x03 (Configuration register): write 0xFC to configure GPIO0–GPIO1 as outputs and GPIO2–GPIO7 as inputs.

Command 0x01 (Output Port register): write desired initial state for GPIO0–GPIO1 (e.g., 0x00 to drive both low at startup, accounting for bits 2–7 being ignored on inputs).

Note: GPIO2–GPIO7 are inputs with external 10 k Ω pull-ups. The PI4IOE5V9554A has fixed internal 100 k Ω pull-ups on inputs that cannot be disabled; they sit in parallel with the externals and are dominated by the 10 k Ω . GPIO0–GPIO1 are configured as outputs.

⚠ This ESP32 features a GPIO Matrix allowing each pin to be connected to any internal peripheral signal (Technical Reference Manual page 115).

However, for better high speed communication performance, direct connections for peripherals will be used unless impossible.
See Table 6.10-1, page 133, Technical Reference Manual

GPIO0, GPIO2, GPIO5, GPIO12 and GPIO15 are strapping pins used during the boot sequence of the ESP32, modifying their default values can make the chip freeze or crash. (ESP32 Datasheet, Table 3-1, page 22)

It is best to leave these pins unconnected or use them as output, since these pins only require a specific value during the boot sequence.

△ DISPLAY SPI :

Since the display uses half-duplex SPI mode (no MISO) so no inputs towards the ESO32, it will be connected using the HSPI peripheral as it has the most strapping pins

SPI.SCK -> GPIO14 (HSPICLK)
SPI.MOSI -> GPIO13 (HSPID)
SPI.CS -> GPIO15 (HSPICS0)
SPI.D/C -> GPIO12
DisplayRST -> GPIO2

SD CARD SPI :

HSPI being used by the display, the other free SPI peripheral is VSPI

SD.CS -> GPIO5 (VSPICS0)
SD.MOSI -> GPIO23 (VSPID)
SD.SCLK -> GPIO18 (VSPICLK)
SD.MISO -> GPIO19 (VSPIQ)
SD.Detect -> GPIO21

SD_CS is connected to the GPIO5 strapping pin with a pull-up resistor but it is not an issue since the default value of this pin is 1, it will not cause any problem for the boot sequence. (ESP32 Datasheet, Table 3-1, page 22)

USB-UART Bridge :

```
UART.RXD -> U0TXD
UART.TXD -> U0RXD
```

I2C Interface :

The I2C peripheral inside the ESP32 has no direct connection pin, so it can be connected to any GPIO using the GPIO Matrix. See ESP32 Datasheet section 4.8.4

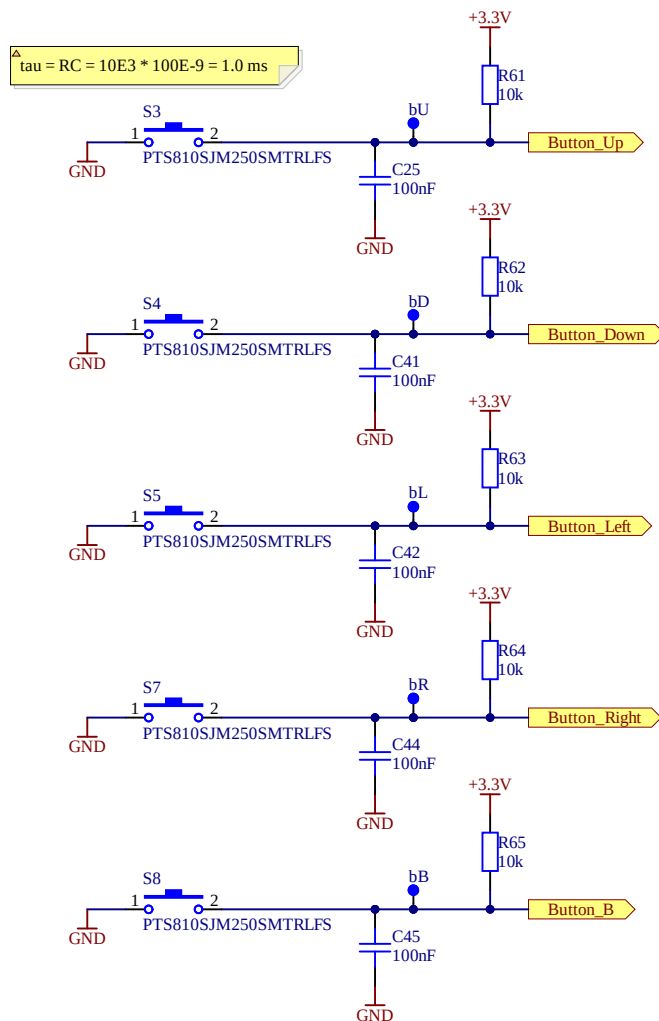
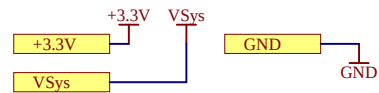
```
I2C.SDA -> GPIO32
I2C.SCL -> GPIO33
```

DAC I2S ·

The I2S peripheral inside the ESP32 has no direct connection pin, so it can be connected to any GPIO using the GPIO Matrix. See ESP32 Datasheet section 4.8.5

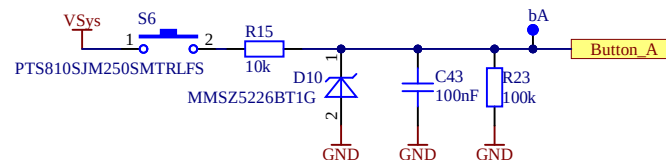
```
I2S.LRCK -> GPIO25
I2S.BCK -> GPIO26
I2S.SD -> GPIO27
```

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Title <title>										
Project Name YDS - TB2026 - Modern Audio Player						Project ID <SGX N°>				
Format	A3	VCS	No	Version	1.0.0	Review	0	Page	5 / 8	
Date	22/07/2026		Author			Yann Dos Santos			Checked by	*
File	Block_ESP32_SchDoc									



Since S6 (Button A) is also used as the Power ON button, it is connected to VSys instead of +3.3V, as the +3.3V is only active once the system has been powered on.

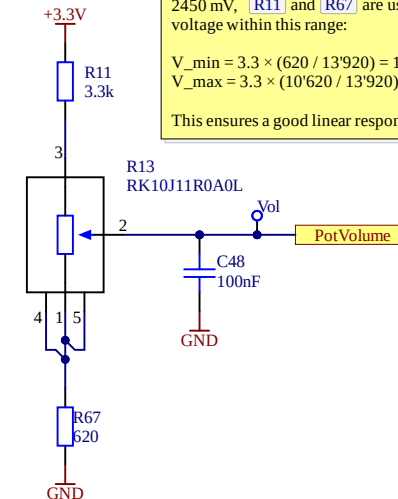
To protect the ESP32 GPIO input, a 3.3V Zener diode is used.



Since the ESP32 linear ADC range is from 150 mV to 2450 mV, R11 and R67 are used to keep the wiper voltage within this range:

$$V_{\min} = 3.3 \times (620 / 13'920) = 147 \text{ mV}$$
$$V_{\max} = 3.3 \times (10'620 / 13'920) = 2.518 \text{ V}$$

This ensures a good linear response of the volume knob



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Title	Block UI
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Project Name
YDS - TB2026 - Modern Audio Player

Project ID	<SGX N°>
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Format	A4	VCS	No	Version	1.0.0	Review	0	Page	6 / 8
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Date	22/07/2026	Author	Yann Dos Santos	Checked by	*
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